INCH-POUND

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DETAIL SPECIFICATION

PACKING, PREFORMED, STRAIGHT THREADED TUBE FITTING BOSS, TYPE II HYDRAULIC (-65°F TO +275°F)

INACTIVE FOR NEW DESIGN AFTER 16 JUNE 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of elastomeric straight threaded tube fitting boss packings for use in 3000 psi (20.68 MPa) hydraulic systems using MIL-PRF-5606 fluid throughout a temperature range of -65°F to +275°F (-54°C to +135°C).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-601	Rubber; Sampling and Testing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance (Inactive for new design)
MIL-PRF-6083	Hydraulic Fluid, Petroleum Base, for Preservation and Operation

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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DEPARTMENT OF DEFENSE STANDARDS

AND10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread
MIL-STD-413	Visual Inspection Guide for Rubber Elastomeric O-Rings
MS21344	Fittings – Installation of Flared Tube, Straight Threaded Connectors, Design Standard for

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1	Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available from www.asme.org or the American Society of Mechanical Engineers International, Three Park Avenue, M/S 10E New York, NY 10016-5990.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this document are available from www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B36/B36M	Plate, Brass, Sheet Strip, and Rolled Bar
ASTM B139/B139M	Rod, Phosphor Bronze, Bar, and Shapes
ASTM D91	Precipitation Number of Lubricating Oils
ASTM D573	Rubber – Deterioration in an Air Oven
ASTM D792	Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1414	Rubber O-Rings
ASTM D2240	Rubber Property – Durometer Hardness

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS-QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
SAE AMS-QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
SAE AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)
SAE AMS4029	Aluminum Alloy Sheet and Plate (2014; -T6 Sheet, -T651 Plate) Solution and Precipitation Heat Treated
SAE AMS6345	Steel, Sheet, Strip, and Plate 0.95 Cr – 0.20 Mo (0.28 – 0.33C) (SAE 4130) Normalized or Otherwise Heat Treated

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SAE ARP-5316	Storage of Elastomer Seals and Seal Assemblies which Include an Elastomer Element Prior to Hardware Assembly
SAE AS5169	Fitting, Port Plug and Bleeder
SAE AS5193	Fitting, Cross, Internal Straight Thread Port
SAE AS5202	Port or Fitting End, Internal Straight Thread, Design Standard
SAE AS33931	Packing, Preformed, Straight Thread Tube Fitting Boss, Hydraulic, -65°F to +275°F

(Copies of these documents are available from www.sae.org or the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The SAE AS33931 packings shall be fabricated from an elastomeric material suitable for use in hydraulic fluid conforming to MIL-PRF-5606. The material shall contain no substances which, when in contact with the fluid will adversely affect the properties of the fluid or packing.

3.3 Design and construction. The design and construction of the packing shall be in accordance with SAE AS33931. The packings shall be made from continuous-circle preforms only.

3.4 Performance. The packing and batch material shall meet the performance requirements specified in Table I performance tests when subjected to the applicable tests.

3.5 Shelf life. Unless otherwise specified, the shelf life limitations for the finished product shall conform to SAE ARP-5316 (see 6.4).

3.6 Finish.

3.6.1 Trimming. Mold flash shall be removed from the packings in such a manner that they conform to the requirements specified herein and in SAE AS33931. The method used to trim the mold flash shall not remove the surface of the packing to a greater extent than would result from a light, nonabrasive buffing.

3.6.2 Surface. All surfaces of packings shall be free from defects, blemishes, or irregularities other than minor pits, raised spots, or tool marks resulting exclusively from mold imperfections (not from dirt in the mold) and which are not greater than .003 inch in height or depth from the packing surface. Such blemishes shall not be grouped closely and shall cover not more than 10 percent of the surface.

3.7 Workmanship. The packing shall be free of unbonded or weak sections caused by internal mold seams, cold shuts, foreign inclusions or other detrimental characteristics. The packing shall be uniform in quality and condition; clean, smooth, and free from foreign materials or defects detrimental to appearance or performance. Major defects as defined in MIL-STD-413 shall be cause for rejection.

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4. VERIFICATION

4.1 Classification of inspection. The inspection of packings shall be classified as follows'

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall consist of all of the inspections of this specification conducted in the order listed under 4.5 and in Table I.

4.2.1 Sampling for first article inspection. First article test samples shall be as specified in Table I and shall include the following:

(a)	24	SAE AS33931, MS3393-3 packings
(b)	4	SAE AS33931, MS3393-4 packings
(c)	4	SAE AS33931, MS3393-8 packings
(d)	4	SAE AS33931, MS3393-12 packings
(e)	58	SAE AS33931, MS3393-16 packings
(f)	3	ASTM hardness discs (.250 in. (6.35 mm) thick, with a 1.0 in (25.4 mm). diameter)

4.2.1.1 Hardness discs. Hardness discs shall be formulated, processed, and cured the same as the accompanying packing. The packings shall represent the manufacturer's production product, particularly with respect to stock preparation, mold design, mold finish, and molding technique.

4.3 Conformance inspection. Conformance inspection shall consist of sampling inspections only, conducted in the order listed in Table II.

4.3.1 Sampling inspection. A random sample shall be selected from each inspection lot or batch in accordance with Table II and shall be subjected to the inspections therein.

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TABLE I. First Article Inspection Tests and Samples 1/

Physical Property Tests	Paragraph Number	Specimen				
Unaged						
Hardness	4.5.2.1	1 ASTM Hardness disc (.250 in. (6.35 mm) thick x 1. 0 in. (25.4 mm) diameter)				
Tensile strength	4.5.2.2	2 SAE AS33931, MS3393-16				
Elongation	4.5.2.3	2 SAE AS33931, MS3393-16				
Specific gravity	4.5.2.4	3 SAE AS33931, MS3393-16				
Permanent set	4.5.2.5	2 SAE AS33931, MS3393-3				
Temperature retraction	4.5.2.6	3 SAE AS33931, MS3393-16				
Corrosion & adhesion	4.5.2.8	12 SAE AS33931, MS3393-16				
Air Aged						
Hardness	4.5.2.1	1 ASTM Hardness disc (.250 in. (6.35 mm) thick x 1. 0 in. (25.4 mm) diameter)				
Tensile strength	4.5.2.2	2 SAE AS33931, MS3393-16				
Elongation	4.5.2.3	2 SAE AS33931, MS3393-16				
Permanent set	4.5.2.5	2 SAE AS33931, MS3393-3				
Oil Aged at 158°F (70°C) Temperature retraction	4.5.2.6	3 SAE AS33931, MS3393-16				
Oil Aged at 275 °F (135°C)						
Hardness	4.5.2.1	1 ASTM Hardness disc (.250 in. (6.35 mm) thick x 1. 0 in. (25.4 mm) diameter)				
Tensile strength	4.5.2.2	2 SAE AS33931, MS3393-16				
Elongation	4.5.2.3	2 SAE AS33931, MS3393-16				
Permanent set	4.5.2.5	2 SAE AS33931, MS3393-3				
Volume change	4.5.2.7	3 SAE AS33931, MS3393-16				
Performance Tests		SAE AS33931, MS3393 Packing				
		-3 Size	-4 Size	-8 Size	-12 Size	-16 Size
Installation stretch at normal temperature	4.5.2.9.1	12				
Installation stretch at -20°F (-29°C)	4.5.2.9.2	6				
Crush	4.5.2.10					9
Extrusion	4.5.2.11					9
Impulse	4.5.2.12		4	4	4	4

1/ All SAE AS33931, MS3393 packings shall be examined in accordance with 4.5.1 before the initiation of first article tests.

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TABLE II. Conformance Inspection

(all specimens unaged except Volume change)				
Inspection	Paragraph Number	Sampling plan per	Inspection level	AQL (percent defective)
Examination of product	4.5.1	ASQ Z1.4	S-4	4.0
Examination of preparation for delivery	4.5.1.1	ASQ Z1.4	S-4	6.5
		Specimen		
Batch 1/, 2/				
(a) Hardness	4.5.3.1	1 ASTM Hardness disc (.250 in. (6.35 mm) thick x 1. 0 in. (25.4 mm) diameter)		
(b) Tensile strength	4.5.3.2	2 SAE AS33931, MS3393-16 packings		
(c) Elongation	4.5.3.3	2 SAE AS33931, MS3393-16 packings		
(d) Specific gravity	4.5.3.4	3 SAE AS33931, MS3393-16 packings		
Finished Product 2/, 3/				
(a) Tensile strength	4.5.3.2	2 SAE AS33931, MS3393-16 packings		
(b) Elongation	4.5.3.3	2 SAE AS33931, MS3393-16 packings		
(c) Specific gravity	4.5.3.4	3 SAE AS33931, MS3393-16 packings		
Process monitoring 2/, 4/				
(a) Permanent set	4,5.3.5	2 SAE AS33931, MS3393-3 packings		
(b) Temperature retraction	4.5.3.6	3 SAE AS33931, MS3393-16 packings		
(c) Volume change	4.5.3.7	3 SAE AS33931, MS3393-16 packings		
(d) Corrosion and adhesion	4.5.3.8	12 SAE AS33931, MS3393-16 packings		
(e) Crush	4.5.2.10	3 SAE AS33931, MS3393-16 packings		

1/ A batch shall consist of the quantity of material compounded at any one time and ready for molding and curing into a fished product. Test specimen shall be molded from each batch and finished with the same state of cure as the finished product.
2/ If a test specimen or sample fails to pass any of these tests, the lot or batch represented by the unit shall be rejected.
3/ Finished product tests shall be performed on unaged specimens or samples from each day's production molded concurrently with other production parts.
4/ Process monitoring test shall be performed at 3 month intervals on unaged specimen molded concurrently with production parts, and using the same production techniques.

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4.4 Inspection conditions.

4.4.1 Ambient (room or normal) temperature and pressure. Unless otherwise specified, tests shall be conducted at an ambient temperature of 75 ± 5°F (24 ± 2.5°C) and at an ambient barometric pressure of 28 to 32 inches (711.2 to 812.8 mm) of mercury. If tests are conducted at different ambient conditions, appropriate correction factors shall be applied to the pressure readings.

4.4.2 Air aging. When required for physical properties tests, air aging shall be in accordance with ASTM D573, except that specimens shall be aged for 166 ± 1 hours at 158 ± 2°F (70 ± 1°C).

4.4.3 Oil aging. When required for physical properties tests, oil aging of test specimens shall be as follows. The fluid and samples to be aged shall be placed in a non-pressurized closed container or reflux condenser. The test fluid shall not be in contact with any part of the container cover. The packings shall be suspended in the fluid by a glass or nichrome wire rack and shall not come in contact with any part of the container. A new batch of fluid shall be used for each aging test. In all cases the ratio of quantity of fluid to the weight of the seals or material being aged shall be 20 ml of fluid for every gram of material. Seals or materials shall be loosely separated in the fluid.

4.4.3.1 Oil aging at 158°F (70°C) for temperature retraction (4.5.2.6) and extrusion (4.5.2.11). Aging (fluid) temperature shall be maintained at 158 ± 2°F (70 ± 1°C) for a period of 166 ± 1 hours.

4.4.3.2 Oil aging at 275°F (135°C) for all other oil aging tests. Aging (fluid) temperature shall be maintained at 275 ± 10°F (135 ± 5°C) for a period of 72 hours.

4.4.4 Test fluid. Unless otherwise specified the fluid used for all applicable tests under 4.5 shall conform to MIL-PRF-5606.

4.5 Inspection methods.

4.5.1 Examination. The sample packings shall be thoroughly examined to determine conformance with this specification and applicable drawings for finish, trimming, marking, cross-sectional dimensions, inside and outside diameters, and all other requirements not covered by tests.

4.5.1.1 Examination of preparation for delivery. The packings shall be examined to ascertain that the preparation for delivery conforms to requirements of Section 5 of this specification.

4.5.2 First article tests.

4.5.2.1 Hardness. This test shall be in accordance with ASTM D2240, using ASTM hardness disc specimens. The test shall be conducted at 75 ± 5°F (24 ± 2.5°C), and the Shore A durometer hardness value for the unaged specimen shall be not less than 75 nor more than 85 points. The hardness value for the air aged specimen (see 4.4.2) shall be within -0 and +5 points of the unaged specimen. The hardness value for the oil aged specimen (see 4.4.3 and 4.4.3.2) shall be within ± 5 points of the unaged specimen.

4.5.2.2 Tensile strength. This test shall be in accordance with ASTM D1414 using two SAE AS33931, MS3393-16 packings each for the unaged, air aged, and oil aged tests. The tensile strength for the unaged specimen shall not be less than 1500 psi (10.34 MPa). The tensile strength values for the air aged specimen (see 4.4.2) and oil aged specimen (see 4.4.3 and 4.4.3.2) shall be at least 80 percent of the value obtained for the unaged specimen.

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4.5.2.3 Elongation. This test shall be in accordance with ASTM D1414 using two SAE AS33931, MS3393-16 packings each for the unaged, air aged, and oil aged tests. The elongation for the unaged specimen shall not be less than 125 percent. The elongations for the air aged specimen (see 4.4.2) and oil aged specimen (see 4.4.3 and 4.4.3.2) shall be at least 70 percent of the value obtained for the unaged specimen.

4.5.2.4 Specific gravity. This test shall be in accordance with ASTM D792, except that three SAE AS33931, MS3393-16 packings shall be used. The specific gravity shall be as determined.

4.5.2.5 Permanent set. Permanent set tests shall be conducted using a suitable tapered cone and two SAE AS33931, MS3393-3 packings each for the unaged, air aged, and oil aged tests. The original inside diameter of the packing shall be measured and the packing then shall be stretched over a tapered cone lubricated with MIL-PRF-5606 fluid until the inside diameter of the sealing surface has been increased 35 percent. The packing shall be held in the stretched state for 10 minutes, removed from the cone, and the new inside diameter determined after waiting 10 minutes. Permanent set (P.S.) shall be calculated as follows:

$$P\,S\,(\%) = \frac{\text{(I.D. stretched packing in inches (mm))-(I.D. unstretched packing in inches (mm))}}{\text{(I.D. unstretched packing in inches (mm))}} \times 100$$

There shall be no evidence of breaking or cracking of the test samples. The permanent set for unaged, air aged (see 4.4.2), and oil aged (see 4.4.3 and 4.4.3.2) samples shall not exceed 3.0 percent.

4.5.2.6 Temperature retraction. This test shall be in accordance with ASTM D1414, except that three SAE AS33931, MS3393-16 packings each for the unaged and oil aged tests shall be used and shall be tested in a test set-up similar to Figure 1. The rate of warming shall be 1°F (.5°C) per minute. The temperature retraction for both unaged and oil aged (see 4.4.3 and 4.4.3.1) samples at 50 percent elongation and 10 percent return (50/10 reading) shall occur at -40°F (-40°C) or lower.

4.5.2.7 Volume change. The volume change of the test sample shall be determined using three SAE AS33931, MS3393-16 packings and shall be conducted in accordance with FED-STD-601, method 6211, after oil aging in accordance with 4.4.3 and 4.4.3.2. The volume of the test sample shall not decrease and shall not increase in excess of 12.0 percent.

4.5.2.8 Corrosion and adhesion. Twelve SAE AS33931, MS3393-16 packings shall be prepared for corrosion testing by prehumidifying the packings in a desiccator or similar humidity chamber, maintained at 92 percent minimum relative humidity and room temperature for 72 hours. Metal plates used shall be polished to a surface roughness of 4 to 16 microinches (.1 to .4 micrometers) Ra finish in accordance with ASME B46.1. The edges shall also be polished to reduce the formation of edge corrosion. Metal plates shall be washed with precipitation naphtha, as specified in ASTM D91, or similar degreasing agent. The metal plates used shall be as follows (a set of seven control plates, one of each of the metals tested shall be included):

Aluminum alloy	SAE AMS-QQ-A-250/4, Condition T62
Aluminum alloy	SAE AMS-QQ-A-250/12, Condition T651 Heat Treated to T73
Aluminum alloy	AMS 4029, 2014 Condition T6
Brass	ASTM B36/B36M, C23000
Phosphorbronze	ASTM B139/B139M, C51000
Steel	SAE AMS6345, Condition N
Steel	Cadmium plated per SAE AMS-QQ-P-416

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The humidified packings and the metal plates shall be immersed in MIL-PRF-6083 fluid and drained to the drip point. The packings and plates shall then be laid together in a stack such that at least two packings contact each specified metal. The stack shall be held together with a force of 20 to 30 pounds and placed in a desiccator which is maintained at 90 to 92 percent relative humidity at room temperature. A separate set of metallic plates shall also be prepared (buffed, cleaned and dipped in MIL-PRF-6083 fluid and drained) and placed in the desiccator in such a manner that the control plates shall not touch each other or any of the packings (dipotassium acid phosphate, K_2HPO_4, when placed in distilled water in sufficient quantity to produce a saturated solution, will maintain approximately 92 percent relative humidity in a sealed desiccator at 68°F (20°C)).

Time of storage for this portion of test shall be 336 hours. No more than 15 minutes shall elapse between the time the test samples are removed from the prehumidifying chamber and placed in the stacked condition in the second humidity chamber.

At the termination of this test, the test samples shall be examined as follows:

(a) The surfaces of the plates that were in contact with the packings shall be inspected for evidence of deposit formation or adhesion of packing material. If deposits or adhesion are evidenced the surfaces of the plates shall be washed in precipitation naphtha. Deposits determined as rubber compounds or elements therefrom, which can be removed by this process and which do not occur on the separate control plates, shall be construed as adhesion. There shall be no evidence of adhesion of the packing material to the metal plates.

(b) If any other marks remain on the surfaces of the plates after step (a), the surfaces shall be lightly polished with a non-abrasive buffing cloth. Any pits or eroded marks remaining after this process shall be construed to be corrosion. Discoloration or staining (marks which do not physically affect the surface of the plates and which easily wash or buff off) shall not be considered detrimental. If any doubt should arise about the presence of pitting, erosion or corrosion on the metal plates, a microscope of approximately 10 to 15-power magnification shall be used to determine the actual condition. There shall be no evidence of severe discoloration, corrosion (pitting), or erosion.

4.5.2.9 Installation stretch.

4.5.2.9.1 Installation stretch at normal temperature. Twelve SAE AS33931, MS3393-3 packings shall be used in this test. Six packings shall be unaged and six shall be air aged (see 4.4.2) unless the result of the elongation test indicated that the material decreased in elongation due to air aging by more than 30 percent, in which case all twelve packings shall be air aged. The packings shall be stretched onto a smooth metallic mandrel of .390 ± .002 inch diameter and allowed to remain in this condition on the .390 inch diameter section of the mandrel for 1 hour at room temperature. There shall be no evidence of breakage or cracking of any of the packings.

4.5.2.9.2 Installation stretch at -20°F (-29°C). The test of 4.5.2.9.1 shall be duplicated at -20 ± 2°F (-29 ± 1°C) except that three each of the unaged and air aged (see 4.4.2) packings shall be used (regardless of change in elongation). The unaged and air aged packings and the threaded mandrel (see Figure 2) shall be held at -20 ± 2°F (-29 ± 1°C) for 24 hours (packings unstretched) prior to stretching on the mandrel. Each packing shall be rapidly pushed from the cone of the mandrel over the threaded .390 inch diameter onto the smooth 0.320 inch diameter undercut. There shall be no breakage of packings in this test and each packing shall decrease in inside diameter to grip the .320 inch diameter undercut in a maximum of 10 seconds after being inserted on the mandrel while still at the specified temperature.

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4.5.2.10 Crush. Nine SAE AS33931, MS3393-16 packings shall be used in this test. Three shall be unaged, three shall be air aged (see 4.4.2), and three shall be oil aged (see 4.4.3 and 4.4.3.2). The packing shall be tested in a test setup similar to Figure 3. The packings shall each be twisted (inside out) one-half turn about the cross section and laid between rigid steel plates which have a surface finish of 40 to 60 microinches (1 to 1.5 micrometers) obtained by sandblasting a commercial grind finish as suggested in Figure 3. The air-aged packings shall be laid together in one layer, the oil aged in a second layer and the unaged in a third layer. The stack shall then be subjected to a constant axial compression load of 16,800 pounds (7630.35 kg), or 5600 pounds (2540.11 kg) for each packing in a layer (number of layers does not affect the applied load). The stack shall be left in the compressed state for 68 hours and then each ring shall be removed and inspected immediately for cracking or disintegration. The packings shall be stored at room temperature in a free condition and the permanent set of seal cross-section determined at the end of 20 hours. No cracking or disintegration shall be evident. The cross sectional area of the packing shall not have been reduced in permanent set more than 25 percent of the "as manufactured" size.

4.5.2.11 Extrusion. Nine SAE AS33931, MS3393-16 packings shall be used for this test. Three shall be used unaged, three air aged (see 4.4.2), and three oil aged (see 4.4.3 and 4.4.3.1). The nine packings shall be installed one each on SAE AS5169-16 fittings, and the fittings shall then be installed in AND10050 bosses of applicable size (SAE AS5193-16 boss fittings may be used). The fittings shall be screwed into the bosses so that an .012 inch feeler gage can be inserted between the flat boss-face and the adjacent face of the SAE AS5169 fitting (this allows .012 inch extrusion gap around the test packings). Hydrostatic pressure of 3000 ± 100 psi (20.68 ± .69 MPa) shall be maintained constantly on the inside of the SAE AS5169 fittings and bosses, and the assemblies shall be maintained at 275 ± 2°F (135 ± 1°C) for 72 hours. During this period there shall be no evidence of excessive extrusion of packing or leakage of fluid, and at the termination of the test the packings shall show no cracking, disintegration or other failure. The packings may remold to the shape of the confining cavity. A small undeteriorated ridge at the extrusion point of the packing shall not be considered cause for rejection.

4.5.2.12 Impulse. Standard SAE AS33931 packings in quantities and sizes specified in Table I shall be used for this test. The unaged packings shall be installed on standard fittings in accordance with MS21344, sheets 2 and 3, which shall, in turn, be installed in a test fixture as shown in Figure 4. The test fixture shall be installed in a test set-up similar to Figure 5. Impulse pressure (see Figure 6) shall be applied at a rate of 35 ± 10 cycles per minute through the range of 0 to 3000 psi (20.68 MPa) with a peak pressure of 150 percent of 3000 psi (20.68 MPa) at the start of each pressure cycle for a total of one million cycles. There shall be no leakage at any time when the packing assembly is subjected to the following schedule (shown graphically on Figure 7):

(a) Conduct a static leak check at 5 psi (.03MPa), 3000 psi (20.68 MPa), and 4500 psi (31.03 MPa).
(b) 50,000 cycles at room temperature, followed by a static pressure leak check at 5 psi (.03MPa), 3000 psi (20.68 MPa), and 4500 psi (31.03 MPa).
(c) 50,000 cycles at 65 ± 5°F (-54 ± 3°C), followed by static leak check.
(d) Static leak check at room temperature.
(e) Static leak check at 275 ± 5°F (135 ± 2.5°C).
(f) 200,000 cycles at 275°F (135°C), followed by static leak check at 275°F (135°C), room temperature and -65°F (-54°C).
(g) 200,000 cycles starting at -65°F (-54°C) and increasing temperature to 275°F (135°C) followed by static leak check at 275°F (135°C), room temperature and -65°F (-54°C).
(h) Repeat paragraph (g) until a total of 1,100,000 impulse cycles have been completed.

4.5.3 Conformance. All specimens used in conformance inspection shall be unaged (except those used for volume change).

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4.5.3.1 Hardness (batch). Test to be same as 4.5.2.1 except that the hardness value shall not vary more than ±5 Shore A durometer points from the value obtained for the unaged specimen in 4.5.2.1.

4.5.3.2 Tensile strength (batch and finished product). Test to be same as 4.5.2.2 except that the tensile strength shall not vary more than ±15 percent from the value obtained for the unaged specimens in 4.5.2.2.

4.5.3.3 Elongation (batch and finished product). Test to be same as 4.5.2.3 except that the elongation shall not vary more than ±20 percent from the value obtained for the unaged specimen in 4.5.2.3.

4.5.3.4 Specific gravity (batch and finished product). Test to be same as 4.5.2.4 except that the specific gravity shall not vary more than ±.02 from the value obtained in 4.5.2.4.

4.5.3.5 Permanent set (process monitoring). Test to be same as 4.5.2.5 except that the permanent set shall not exceed 5.0 percent.

4.5.3.6 Temperature retraction (process monitoring). Test to be same as 4.5.2.6 except that the temperature retraction shall not vary more than ±2°F (±1°C) from the value obtained for the unaged sample in 4.5.2.6.

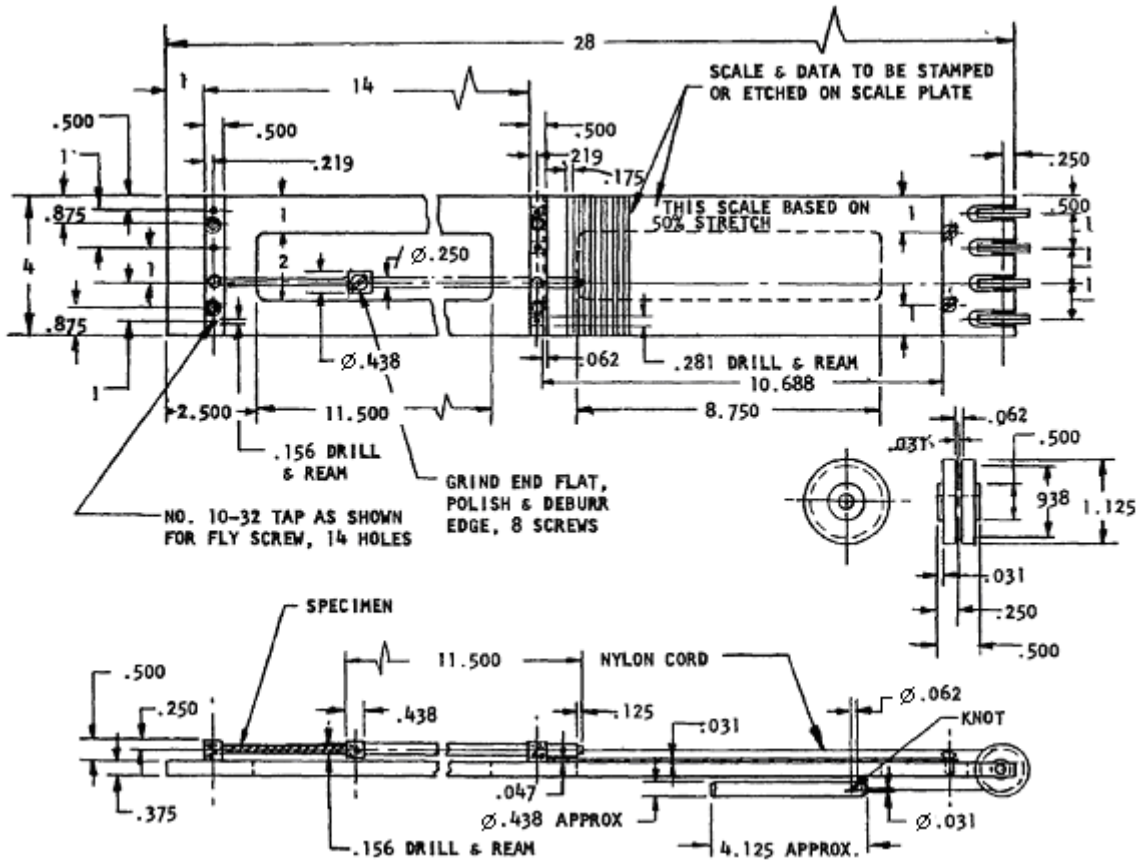
4.5.3.7 Volume change (process monitoring). Test to be same as 4.5.2.7 except that the volume change of the oil aged packing shall be within ±3 percent of the value obtained in 4.5.2.7, and the test sample shall not shrink.

4.5.3.8 Corrosion and adhesion (process monitoring). Test to be same as 4.5.2.8 except that the storage time for the test samples in the desiccator for the final portion of the test shall be 96 hours instead of 336 hours.

5. PACKAGING

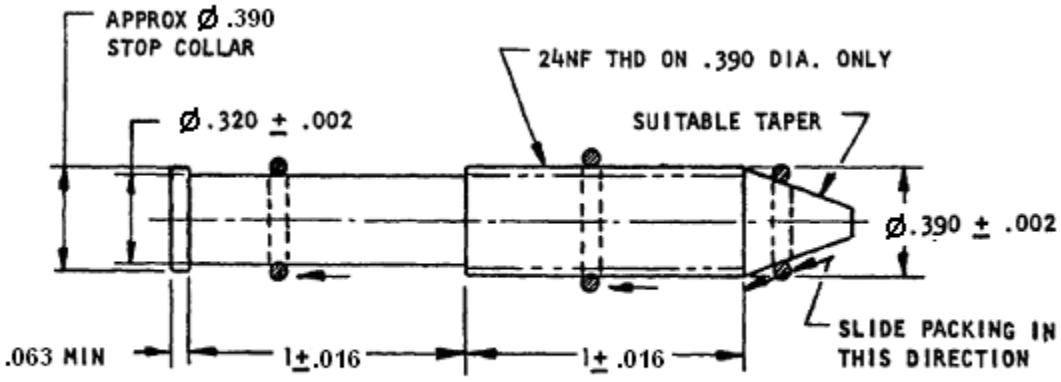
5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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FINISH: PLATE WITH COPPER THEN HARD CHROME ALL OVER.
DIMENSIONS IN INCHES

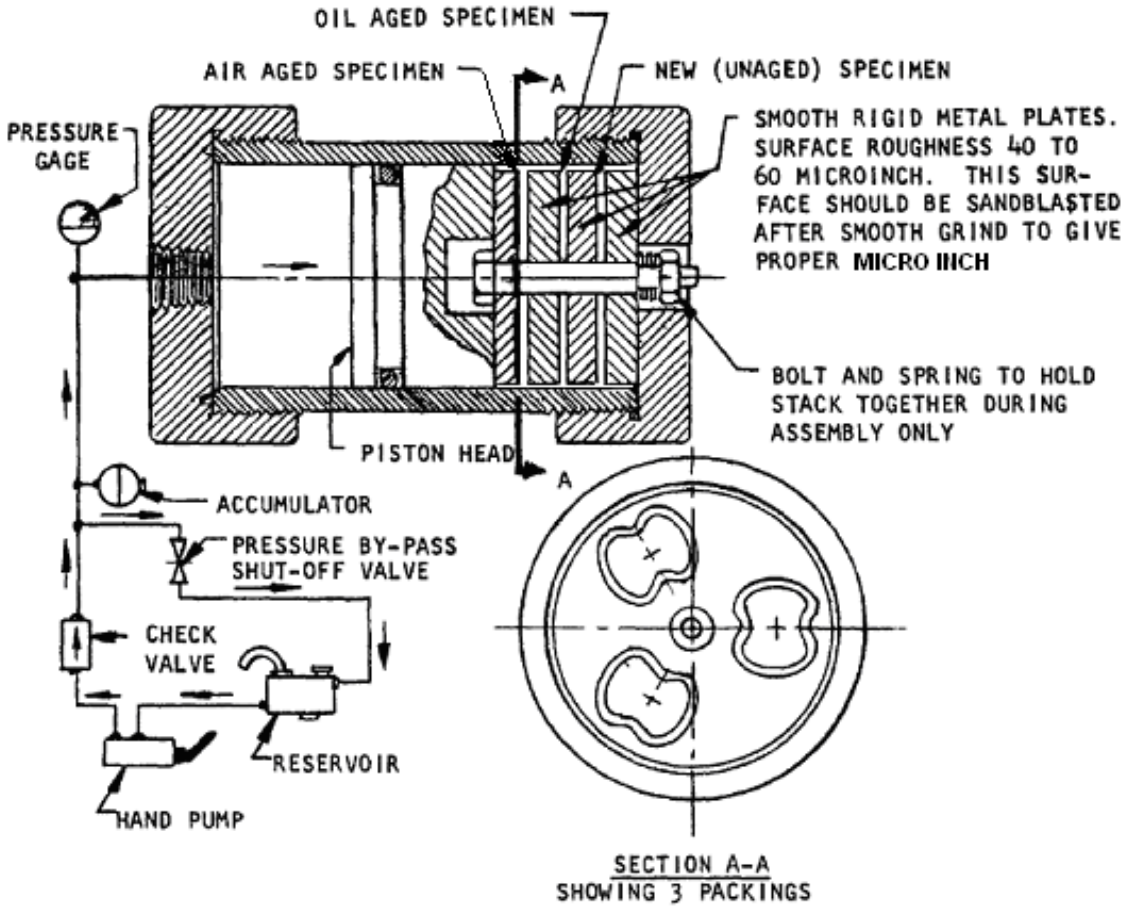
Figure 1. Temperature retraction apparatus.



DIMENSIONS IN INCHES

Figure 2. Installation stretch test mandrel.

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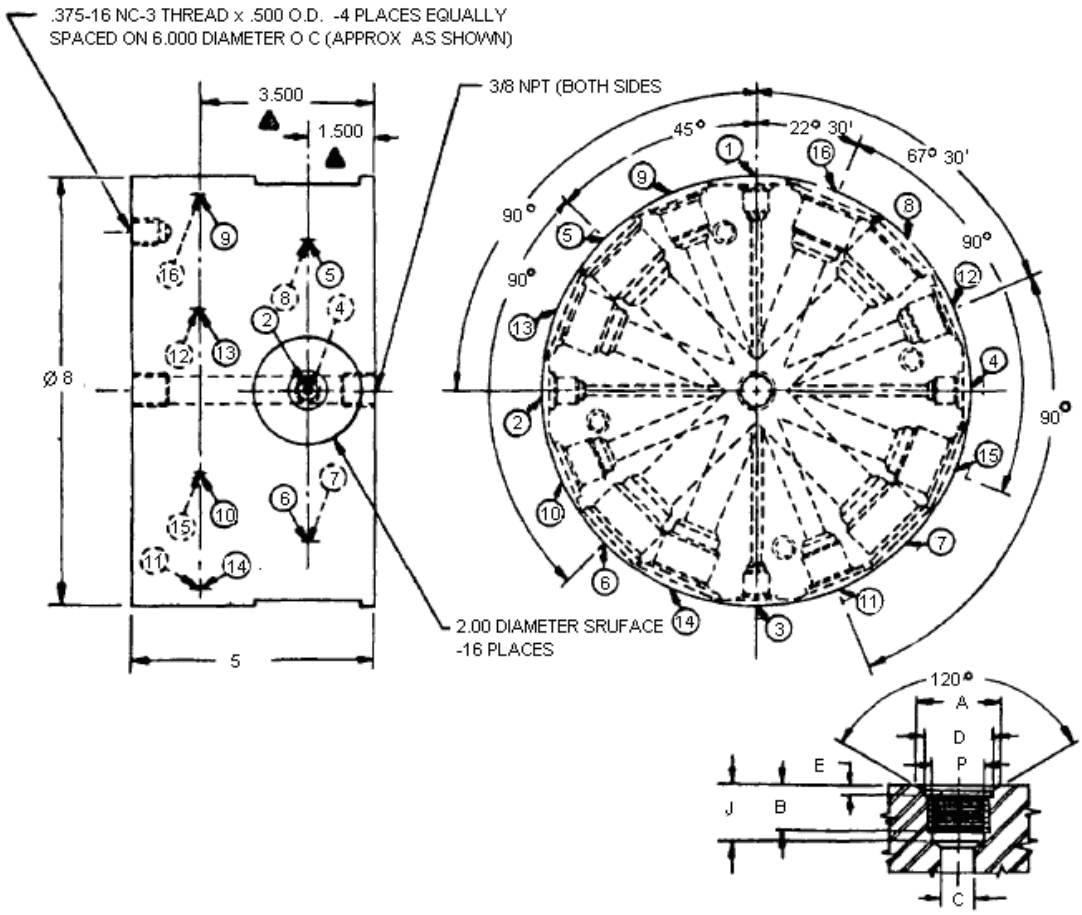


SUGGESTED METHOD OF MAKING CRUSH TEST. HYDRAULIC RIG SHOWN MAY BE MECHANICAL PROVIDED AXIAL CRUSHING LOAD CAN BE ACCURATELY DETERMINED AND MAINTAINED. HYDRAULIC PRESSURE TO COMPRESS PACKING $= P = \frac{F \times N}{A}$

WHERE	P	=	PRESSURE, PSI
AND	N	=	NO. SAE AS33931 MS3393-16 PACKINGS IN EACH LAYER
	A	=	PISTON HEAD AREA IN SQUARE INCHES
	F	=	5600 LBS. FOR SAE AS33931 MS3393-16 PACKINGS

Figure 3. Crush test setup.

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AND10050 BOSS OR SAE AS5202

NOTE:

1. FINISH: COMMERCIAL CAD PLATE LESS THAN .0002 MAX (PER SURFACE)
2. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED

	BOSS	THREAD T	A +.002 -.000	B (MIN)	C (MIN)	D +.001 -.000	E +.001 -.000	J (MIN)	P (PILOT)
▲	(1) and (2)	.438-20 NF-3	.575	.438	.172	.458	.089	.594	.385
	(3) and (4)		.562	.438	.172	.454	.075	.594	.385
▲	(5) and (6)	.313-12 NF-3	1.500	.688	.844	1.340	.139	.938	1.225
	(7) and (8)		1.487	.688	.844	1.336	.125	.938	1.225
▲	(9) and (10)	.250-16 NF-3	.888	.562	.391	.773	.108	.718	.686
	(11) and (12)		.875	.562	.391	.769	.094	.718	.686
▲	(13) and (14)	1.063-12 NF-3	1.247	.688	.609	1.090	.139	.906	.975
	(15) and (16)		1.234	.688	.609	1.086	.125	.906	.975

Figure 4. Impulse test fixture.

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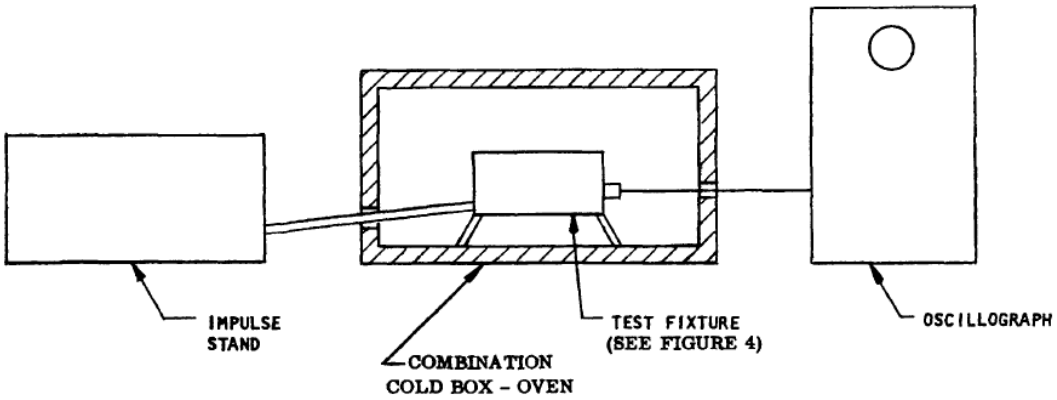
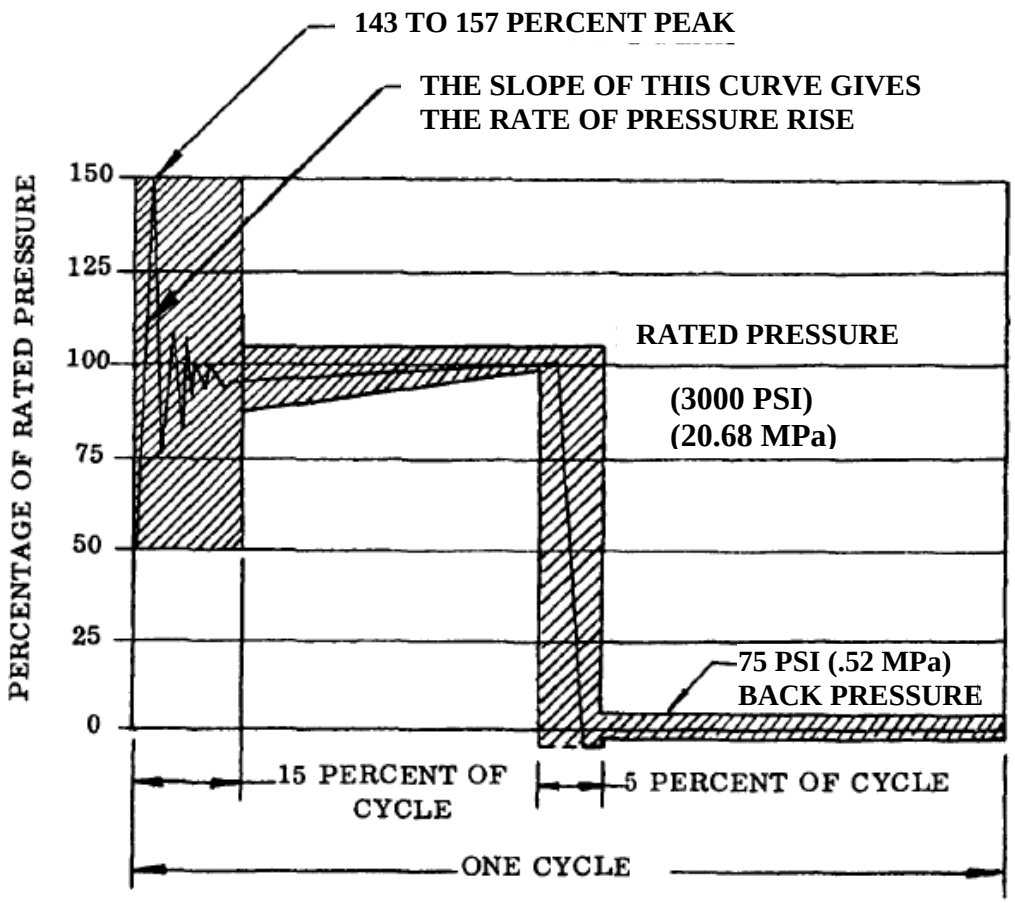


Figure 5. Impulse test setup.

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The curve shown above is the approximate pressure time cycle determined to be of proper severity for impulse testing of hydraulic packings. Although it is mandatory only that pressure peak rises to 143 to 157 percent of the rated pressure (3000 psi) (20.68MPa) at some point prior to leveling off at rated pressure, it is considered highly desirable that the pressure time curve be confined to the shaded area indicated. One very desirable benefit to be gained in this manner is that results of tests performed on different test machines will be more nearly comparable.

Figure 6. Graph of pressure impulse cycle for impulse test.

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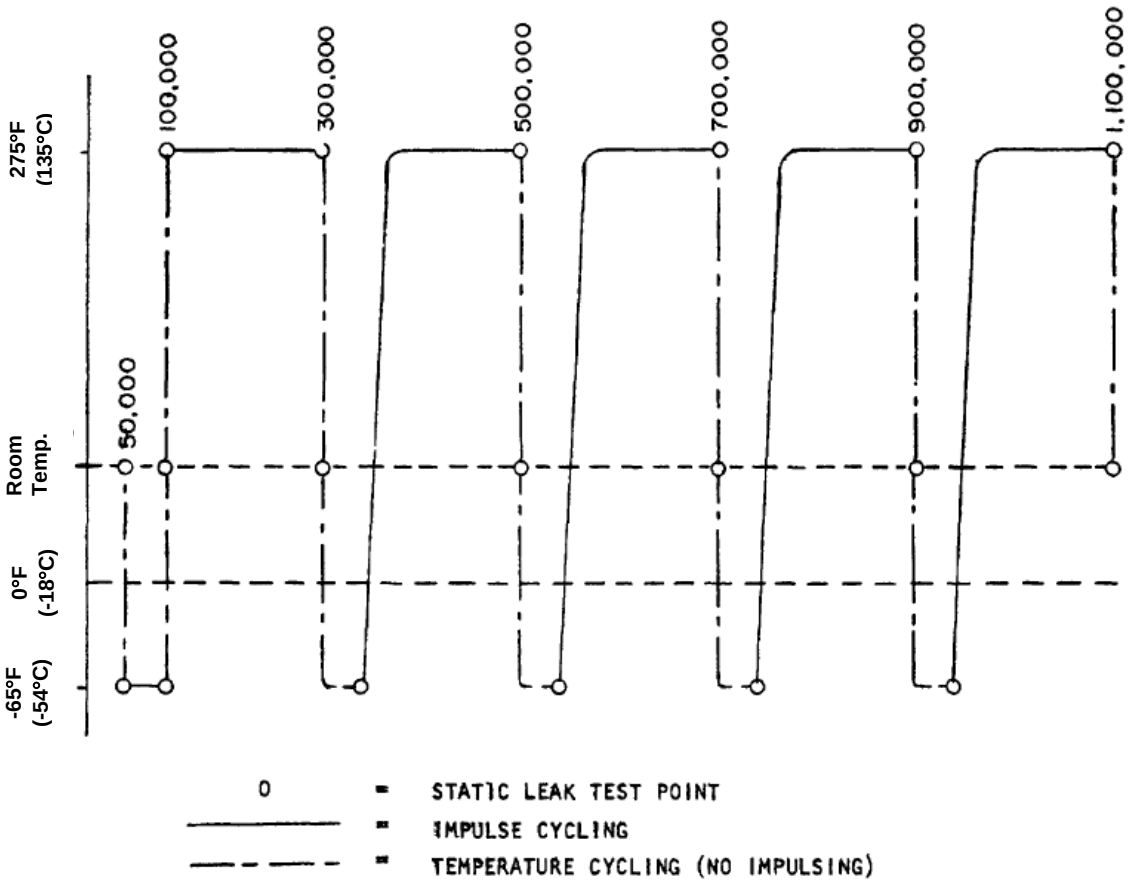


Figure 7. Impulse cycling schedule.

6. Notes.

(This section contains Information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The packings are intended for use in type II class 3000 psi (20.68 MPa) aircraft hydraulic systems using MIL-PRF-5606 to seal standard tube fittings in bosses as indicated in MS21344 (sheets 2 and 3).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. MS part number (see applicable MS sheet).
c. If first article samples are required (see 3.1).
d. Packaging requirements (see 5.1).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the first article (s) should be a preproduction sample, an initial production sample, a first production item or a standard production item from the contractor's current inventory; the number of samples to be inspected as specified in 4.2; and (when applicable) the specific tests to be performed on each sample. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a material quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoD 4140.27-M, *Shelf-life Management Manual*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for Shelf-life management may be obtained from DoD 4140.27-M, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Services and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.hq.dla.mil/.

6.5 Subject term (key word) listing.

Aircraft
Cadmium
Elastomeric
Hydraulic systems
Rubber seal
Type II temperature

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Preparing activity:
DLA – IS

(Project 5330-2009-005)

Review activities:
Army – MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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